

MARBLE Metamorphosed limestone that is very hard and can be polished for use in construction and sculpture. See also *limestone*, *thermal metamorphism*.

MARSH A wetland dominated by plants such as grasses and reeds. Marshes are characterized by regular inundations, which can be of seawater or freshwater. See also *salt marsh* (panel), *swamp*.

MASS MOVEMENT The downslope movement of mud, soil, or rocks, as in a landslide. This process often takes place when soil and other surface debris are saturated with water. See also *transport*.

MASSIF A well-defined mountain mass or group of connected mountains whose rocks and landforms tend to be similar.

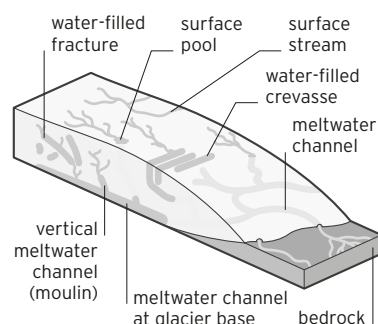
MASSIVE A term describing a rock without any obvious structures (such as bedding planes) or a mineral that is not visibly crystalline.

MATRIX The fine-grained material in which larger grains (in sedimentary rocks) or crystals (in igneous rocks) are embedded. See also *groundmass*.

MEANDER A loop in the course of a river. Meandering rivers gradually change their routes, as erosion takes place on the outside of the loop and deposition on the inside. See also *river valley*.

MEDIAL MORaine A ridge of sediment running along the center of a glacial valley, and parallel with its sides. Medial moraines often form when two glaciers converge and two of their lateral moraines merge. See also *confluence*, *lateral moraine*, *moraine* (panel).

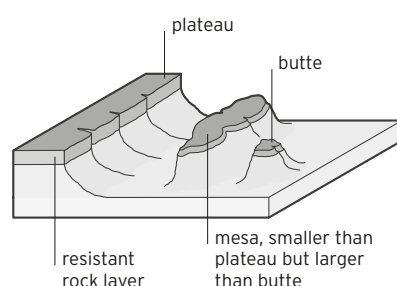
MELTwater Water, from melted snow and ice, that flows within or from the snout of a glacier.



MELTwater CHANNEL ▲

A channel cut by melted snow or ice flowing beneath, through, or near a glacier. The channel and its deposits usually remain after the glacier has retreated. See also *meltwater*.

MESA ▼ A “table-topped,” steep-sided hill. Mesas are characterized by horizontal strata, capped by resistant top layers.



MESOSPHERE The layer of Earth’s atmosphere between the stratosphere and thermosphere. It ranges from about 30 to 50 miles (50 to 80km). See also *stratosphere*, *thermosphere*.

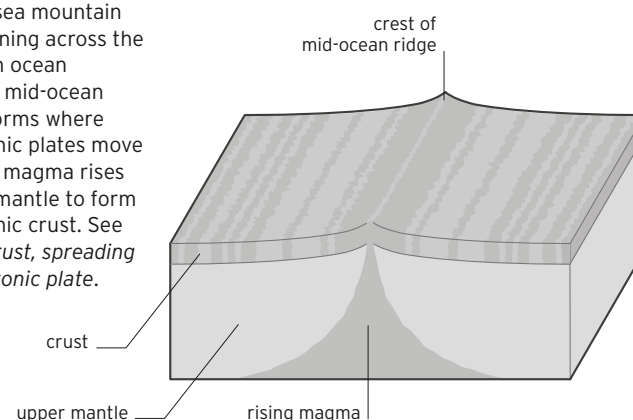
METAMORPHIC ROCK A rock that, under the influence of heat, pressure, or a combination of both, has had its texture or mineral makeup altered.

METAMORPHISM The processes—mainly heating and extreme pressure—by which rocks are transformed under ground. See also *contact metamorphism*, *dynamic metamorphism*, *regional metamorphism*, *thermal metamorphism*.

MICROCLIMATE The distinctive longterm weather conditions of a particular small place, such as a hilltop or a small valley.

MID-OCEAN RIDGE

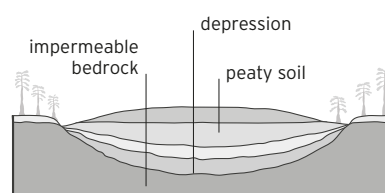
An undersea mountain range running across the floor of an ocean is called a mid-ocean ridge. It forms where two tectonic plates move apart and magma rises from the mantle to form new oceanic crust. See *oceanic crust*, *spreading ridge*, *tectonic plate*.



MID-OCEAN RIDGE See panel, above.

MINERAL A naturally occurring inorganic material that has a well-defined chemical composition. Most rocks are a mixture of several different minerals.

MIRE ▼ An area of swampy ground that retains wet, peaty soil. A mire often forms in a depression in impermeable rock.



MONSOON Refers to a pattern of seasonal winds, especially in southern Asia. The term is also used to refer to the heavy rains these winds bring during the rainy, or monsoon, season. The southwest monsoon brings heavy rain to India, usually between June and September. These winds blow from one direction for about half the year, and from the opposite direction for the other half.

MONTANE Relating to mountains—for example, montane climate or montane fauna.

MORaine See panel, below.

MORaine

Ridges of silt, sand, gravel, and rock debris resulting from glacial action are called moraines. These features often remain after a glacier retreats. Medial moraines form where two glaciers converge, lateral moraines form along the sides of a glacier, and terminal moraines form at the furthest extent of the ice.

